

Efficient Domain-Incremental Deepfake Detection via Difficulty-Aware Curriculum Learning

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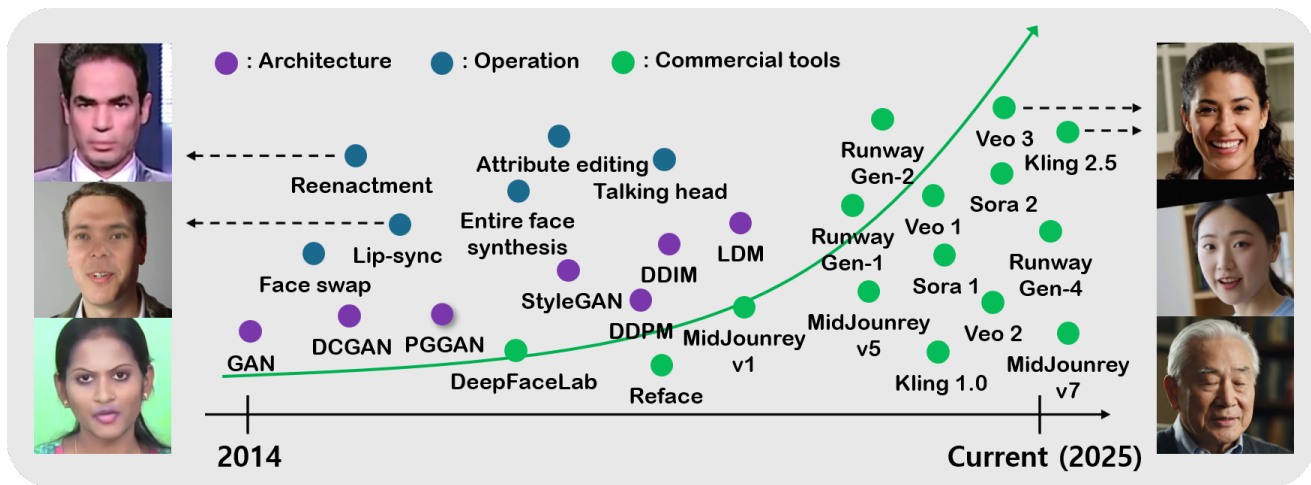


Figure 1. Illustration of the rapid growth of generative models. Starting from early generative models and basic face-swapping techniques, a wide variety of architectures and deepfake generation methods have emerged. In recent years, the trend has accelerated sharply with the rise of high-quality commercial generative video models.

Abstract

Deepfake detection models trained on a single domain often struggle to generalize to others with large domain gaps. Considering that new generative models are rapidly emerging, it is highly challenging to expect strong performance on new forgeries without continual updates, yet continual learning remains underexplored in this field. We propose a domain-incremental training framework that leverages lightweight, difficulty-aware caches and curriculum learning for efficient adaptation. Incoming samples are selectively stored into easy, medium, and hard caches based on entropy, feature distance, and expressiveness. These caches serve to preserve prior knowledge, adapt to novel samples, and bridge domain gaps. Curriculum learning is applied over the caches using an easy-to-hard sampling schedule combined with gradually increasing augmentation strength to maximize generalizability. In addition, contrastive loss is incorporated to cluster easy samples

to retain previous knowledge while push hard ones apart to enhance feature robustness. Our framework demonstrates strong performance on recent deepfake benchmarks, achieving effective knowledge retention, domain adaptation, and improved generalization, even when trained with only 3% of the new data.

1. Introduction

Separating synthetic images from authentic ones is becoming increasingly important as generative models evolve rapidly [9, 10, 39, 40]. While these models are widely used for constructive purposes in various fields including art, media, and advertisement, they are also exploited by malicious actors to spread false information, create illegal pornography, and conduct online scams. Despite the remarkable pace of generative technology, research on detecting such forgeries has often struggled to keep pace. Existing bench-

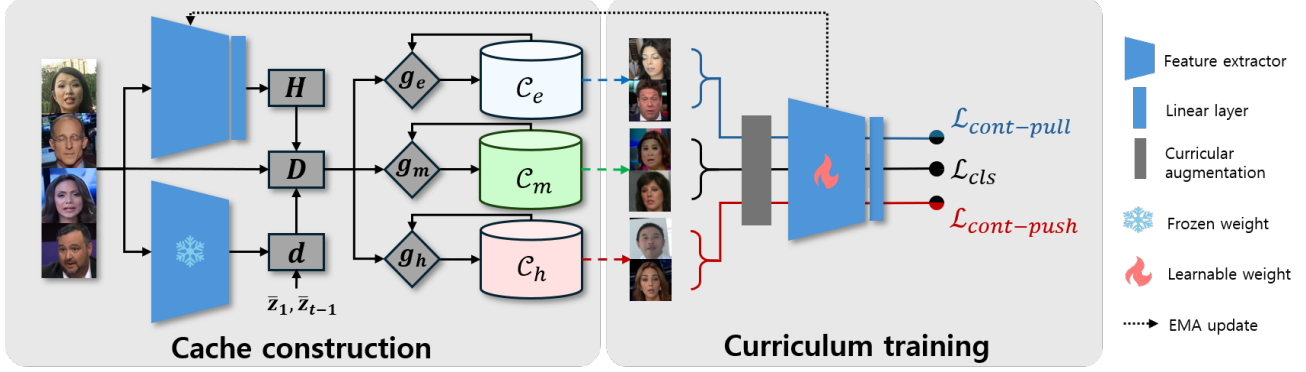


Figure 2. The proposed method is composed of two primary components: cache construction and curriculum training. First, the difficulty D of each incoming sample is estimated by integrating its entropy H and feature distance d . Samples are subsequently then allocated to easy, medium, or hard caches via gating functions (g_e, g_m, g_h) that incorporate both D and intra-cache diversity. Finally, the curriculum training applies an easy-to-hard strategy with increasing augmentation intensity, while contrastive loss is used on easy and hard caches via pull and push operations, respectively.

marks such as FaceForensics++ [41], Celeb-DF [25], and DFDC [8] do not account for recently developed generation techniques based on diffusion models. Although preserving the knowledge derived from these established datasets remains essential for detecting earlier types of forgeries, relying exclusively on them hinders the ability to achieve robust detection performance in emerging domains. Furthermore, many recent studies aimed at improving generalization in deepfake detection [13, 18] still depend heavily on these conventional datasets, therefore often fail to identify fake images generated by modern methods which often exhibit distinct underlying characteristics. As Yan *et al.* [54] highlight, training on data with limited diversity and realism can lead to a significant discrepancy between real-world content and the representations learned by forgery detectors.

Generative methods have evolved continuously from Variational Autoencoders [20] and GANs [11], to the recent diffusion models [15]. While recent studies on generative models [7, 16, 48] and commercial applications [6, 22, 33] have demonstrated impressive quality, detection research often remains underdeveloped, largely relying on outdated datasets. Training solely on datasets composed of outdated generation methods can lead to poor performance in detecting fake images produced by state-of-the-art techniques. Even contemporary datasets face inherent limitations, as they are not able to cover all emerging synthesis methods and inevitably become obsolete like their predecessors as new technology arises. Furthermore, the diversity of forgery types such as face swapping, reenactment, and inpainting introduces another axis of diversity that presents an additional challenge to generalizability. A model trained on a static set of forgery types without continual updates is unlikely to generalize well when exposed to newly introduced manipulation techniques. Due to the limited domain diver-

sity in existing training datasets, there is a substantial risk that models will fail to detect forgeries when encountering a significant domain gap. As demonstrated in our experiments, a network trained exclusively on images from latest commercial tools (*e.g.*, Veo [6] and Kling [22]) achieves strong in-domain performance but suffers significant degradation in out-of-distribution scenarios. While a straightforward approach of training a model with a sufficiently large amount of diverse forgery data seems ideal, accumulating all such data is highly inefficient in terms of both storage and computation.

To address the aforementioned challenges, we explore Domain Incremental Learning (DIL) [49, 50, 57] as a viable solution, which enables models to efficiently adapt to newly emerging domains. Updating a model previously tuned on specific data with newly acquired datasets typically causes performance degradation on the previous data. This phenomenon is known as catastrophic forgetting and has been widely studied in the continual learning literature, with mitigation strategies such as weight regularization [21, 26] and experience replay using stored samples [30]. While some prior works in deepfake detection [19, 34, 53] have explored continual training, our work focuses on efficient adaptation to new domains in an incremental learning manner. The proposed framework does not require any domain labels of incoming data; instead, it leverages a compact replay set, maintained independently of the number of previous domains, and training samples adaptively selected from the current domain.

Generalization is another critical factor to consider, as training on data from all possible domains is often infeasible. A reliable detection model must perform robustly under some degree of domain shift, for instance, when encountering images generated by the same forgery method but in-

volving different identities, or those created via similar generative approaches. While cross-domain evaluations using FaceForensics++ [41], Celeb-DF [25], and DFDC [8] are standard practice, they often fail to account for the patterns of recently developed forgery methods. Our experiments on the DF-40 dataset [54] demonstrate that our framework enhances generalization to these emerging techniques, outperforming the naïve strategy of processing all new data without distinction.

In this work, we focus on three essential objectives for reliable deepfake detection: *i*) efficient adaptation to new data, *ii*) performance retention of previous knowledge, and *iii*) generalizability to unseen domains. A model that fails in any of these aspects is unlikely to be effective for real-world deployment. To meet these requirements and address the challenges of adaptation, retention, and generalization, we propose a novel curriculum-driven, cache-based training framework in a domain-incremental learning setting. Our approach employs lightweight, difficulty-aware caches to store and sample representative training examples based on entropy, feature distance, and expressiveness. These caches, combined with a curriculum strategy [1, 12, 51], enable the model to efficiently adapt to new data. Since our method operates at the input level without modifying the underlying network architecture or feature space, it can be seamlessly integrated with various architectures. Furthermore, it supports continual updates to counter emerging forgery methods, allowing the resulting model to serve as a robust pre-trained detector for future scenarios. Figure 1 illustrates the rapid expansion of generative models and deepfake creation techniques accessible to non-experts, motivating the need for our domain-incremental deepfake detection framework.

Our key contributions are summarized as follows:

- We propose an efficient training framework that utilizes difficulty-aware caches to store representative key samples from new domains. This selection process is governed by a gating mechanism that balances knowledge retention with adaptation to new data.
- We introduce a curriculum-based training strategy involving three difficulty levels (easy, medium, and hard). This approach enables the effective use of limited data while progressively enhancing domain generalization and model robustness.
- Our framework achieves promising performance on recent deepfake detection benchmarks, showing strength in knowledge retention and cross-domain generalization even when trained with only 3% of new data.

2. Related Work

2.1. Deepfake Detection

Deepfake detection aims to distinguish AI-generated images from authentic ones. Given that facial deepfakes are

the primary target of malicious use, research in this field has predominantly focused on human faces. Early approaches utilized spatial cues [24, 41], while more recent studies have leveraged frequency-domain information [18, 31, 37, 45, 46, 56] to capture subtle artifacts. Furthermore, data augmentation in both spatial domain [42] and spatio-temporal shifts [27] have been extensively explored. To enhance model generalizability, Lin *et al.* [27] and Song *et al.* [44] recently introduced curriculum learning to the deepfake detection pipeline. Our work diverges from these existing methodologies by proposing a unified framework that integrates curriculum-based training with domain-incremental learning. The synergy demonstrates strong robustness on recent diffusion-based forgery images. Moreover, our approach incorporates lightweight difficulty-aware caches, facilitating efficient adaptation to emerging domains without significant computational overhead.

2.2. Domain Incremental Learning

Domain Incremental Learning (DIL), a subcategory of continual learning, involves sequentially training a model on data from previously unseen domains while maintaining a fixed task and output space. A primary challenge in DIL is catastrophic forgetting, where the model’s performance on prior domains degrades as it adapts to new ones. To address this, existing DIL methods are generally categorized into replay-based and replay-free approaches.

Replay-based methods preserve knowledge by storing and re-incorporating data from preceding domains. For instance, Chaudhry *et al.* [2] introduced class-specific anchors to enhance memory retention, while Pan *et al.* [34] employed a strategy combining knowledge distillation from a frozen teacher model with the selective preservation of past samples. Conversely, replay-free approaches aim to mitigate forgetting without the need for data storage. Early work by Kirkpatrick *et al.* [21] utilized weight regularization based on parameter importance. More recently, Wang *et al.* [49] proposed a framework that leverages class text features to construct multi-grained class-representative vectors. Similarly, Zhou *et al.* [57] focused on consolidating feature extractors and calibrating classifiers to align aggregated knowledge in an exemplar-free setting, and Wang *et al.* [50] suggested selective parameter updates during fine-tuning to bypass the need for rehearsal. Additionally, Xiong *et al.* [53] introduced domain-specific learnable adapters to facilitate cross-domain knowledge transfer. Specifically within the context of deepfake detection, Kim and Woo [19] leveraged a teacher-student architecture coupled with representation loss to ensure continuous learning.

2.3. Curriculum Training

Training a neural network with a carefully designed curriculum has been shown to accelerate convergence and im-

prove generalization [1]. A common framework in curriculum learning involves defining a difficulty measuring function to evaluate the complexity of training samples, and a pace function that determines which samples should be included in training at a given stage. Typically, easy samples are prioritized in the early training stages while harder samples are progressively introduced as training evolves.

Kumar *et al.* [23] introduced self-paced learning, where the model automatically estimates the difficulty of training samples by solving a bi-convex optimization problem. Sinha *et al.* [43] proposed controlling the smoothness of intermediate features, gradually reducing noise from untrained parameters by annealing the standard deviation of a Gaussian kernel over time. Lin *et al.* [27] applied an easy-to-hard curriculum by adjusting the proportion of augmented images, whereby a detection model and an augmentation-controlling policy model are trained alternately. Lastly, Song *et al.* [44] defined sample difficulty based on a combination of pre-trained image quality assessment and the detection model’s own output, enabling the model to gradually focus on more challenging samples throughout training.

3. Method

In this section, we describe our framework, which consists of data selection and curriculum-based training that leverages selected samples. Let $\mathcal{X}_t = \{(x_t, y_t)\}_{i=1}^N$ denote a dataset containing N pairs of images x_t and labels y_t . Our objective is to efficiently adapt the weights θ_t of a model f , previously trained on $\mathcal{X}_{t-1}$, to a new domain represented by $\mathcal{X}_t$ for $t \geq 1$ while preventing the forgetting of prior knowledge and maximizing generalizability. Figure 2 depicts our full framework of cache-based curriculum training.

3.1. Difficulty-Aware Cache Sampling

3.1.1. Difficulty measurement

We take advantage of using a cache, an adaptively selected subset of the full dataset, for training to reduce computational burden compared to using the entire dataset [55]. For adaptive selection, difficulty and expressiveness are defined to identify samples that contribute most effectively to the training process. The difficulty score D_t^x of a sample x_t is estimated by a combining entropy and feature distance, reflecting both the current state of the model and the informativeness of the new sample:

$$D_t^x = \alpha \cdot H_t^x + (1 - \alpha) \cdot d_t^x, \quad (1)$$

where H_t^x is the entropy of the output probabilities, d_t^x is the distance between the incoming feature and historical the historical feature distribution, and α is a balancing coefficient.

The entropy H_t^x is computed using an exponential moving average (EMA)-updated model f_ϕ for the incoming

Algorithm 1 Pseudo code for cache construction

```

1: Input: Anchor vector  $\bar{z}_1$ , adaptive vector  $\bar{z}_{t-1}$ , current
   data  $\mathcal{X}_t$ , sampling ratio  $r$ , feature extractor of previous
   task  $h_\theta$ , EMA model  $f_\phi$ , thresholds  $\tau_e, \tau_m, \tau_h$ , gating
   functions  $g_e, g_m, g_h$ 
2: Output:
   Easy cache  $\mathcal{C}_e$ , medium cache  $\mathcal{C}_m$ , hard cache  $\mathcal{C}_h$ 
3: Initialize caches:  $\mathcal{C}_e \leftarrow \phi, \mathcal{C}_m \leftarrow \phi, \mathcal{C}_h \leftarrow \phi$ 
4: Sample subset:  $\mathcal{X}'_t \leftarrow \mathcal{X}_t$ 
5: for image  $x$ , label  $y$  in current dataset  $\mathcal{X}'_t$  do
6:   Compute Entropy:  $H_t^x = \sum -f_\phi(x) \log(f_\phi(x))$ 
7:   Compute feature distance:
      $d_t^x = \frac{1}{2}(\|\bar{z}_1 - z_t^x\|_2 + \|\bar{z}_{t-1} - z_t^x\|_2)$ 
8:   Measure difficulty:  $D_t^x = \alpha \cdot H_t^x + (1 - \alpha) \cdot d_t^x$ 
9:   Update thresholds:  $\tau_e, \tau_m, \tau_h \leftarrow EMA(D_t^x)$ 
10:  for  $\mathcal{C}, g, \tau$  in  $[\mathcal{C}_e, \mathcal{C}_m, \mathcal{C}_h], [g_e, g_m, g_h], [\tau_e, \tau_m, \tau_h]$ 
    do
11:    Measure expressiveness:
       $E_{\mathcal{C},x} = 1 - \frac{1}{|\mathcal{C}|} \sum \frac{h_\phi(x) \circ z_{\mathcal{C}}}{\|h_\phi(x)\| \cdot \|z_{\mathcal{C}}\|}$ 
12:    if  $g(\mathcal{C}, x, \tau) = 1$  then
13:      Update cache:  $\mathcal{C} \leftarrow x$ 
14:    end if
15:  end for
16: end for
17: Return results:  $\mathcal{C}_e, \mathcal{C}_m, \mathcal{C}_h$ 

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sample x_t . The feature distance d_t^x is measured as the distance between the historical feature distribution and the feature z_t^x of the incoming sample. The historical distribution is represented by an anchor vector $\bar{z}_1$ and an adaptive vector $\bar{z}_{t-1}$, while z_t^x is extracted by the feature extractor h_θ trained on previous tasks. This process can be formulated as follows:

$$H_t^x = - \sum f_\phi(x_t) \log(f_\phi(x_t)), \quad (2)$$

$$d_t^x = \frac{1}{2}(\|\bar{z}_1 - z_t^x\|_2 + \|\bar{z}_{t-1} - z_t^x\|_2), \quad (3)$$

$$z_t^x = h_\theta(x_t), \quad (4)$$

where $\bar{z}_1$ is the mean feature of the first task acting as an anchor, and $\bar{z}_{t-1}$ is the EMA-updated mean feature from tasks 1 to $t - 1$. Using $\bar{z}_1$ as a fixed anchor and $\bar{z}_{t-1}$ as a cost-efficient adaptive memory provides stability when measuring the divergence of x_t from prior knowledge. The resulting difficulty score D_t^x serves as an effective criterion for cache construction.

3.1.2. Difficulty-Aware Cache construction

We propose a novel cache design comprising three distinct difficulty-aware caches: $\mathcal{C}_e$, $\mathcal{C}_m$, and $\mathcal{C}_h$, which store easy, medium, and hard samples from the new domain, respectively. This difficulty-aware caching strategy is tailored for

domain incremental learning: the easy cache $\mathcal{C}_e$ mitigates catastrophic forgetting, the hard cache $\mathcal{C}_h$ facilitates adaptation to new information, and the medium cache $\mathcal{C}_m$ bridges the gap between the two.

We introduce gating functions g_e , g_m , and g_h to determine sample inclusion based on difficulty and expressiveness. Rather than employing a naïve approach with fixed thresholds which requires sensitive tuning affected by both the model and the training data, our gating functions utilize adaptive thresholds updated via a moving average of the difficulties of incoming samples.

Expressiveness E is designed to enhance the diversity of samples within the cache, complementing the difficulty score D . We measure E by calculating the average cosine similarity at the feature level between an incoming candidate sample x and the samples already stored in the cache $\mathcal{C}$. To normalize the expressiveness score within the range $[0, 1]$, the computation is defined as:

$$E_{\mathcal{C},x} = 1 - \frac{1}{2|\mathcal{C}|} \sum_{z_{\mathcal{C}} \in \mathcal{C}} (\cos(h_{\phi}(x), z_{\mathcal{C}}) + 1), \quad (5)$$

where $\cos(\cdot, \cdot)$ denotes the cosine similarity, h_{ϕ} is the feature extractor of the EMA model, and $z_{\mathcal{C}}$ represents the feature stored in a cache $\mathcal{C}$. For notational brevity, we omit the task subscript t in this subsection.

The gating functions are then defined as follows:

$$g_e(\mathcal{C}_e, x, \tau_e) = \begin{cases} 1, & \text{if } D^x < \tau_e \text{ and} \\ & D^x - \beta \cdot E_{\mathcal{C},x} < \delta_e, \\ 0, & \text{otherwise} \end{cases}, \quad (6)$$

$$g_m(\mathcal{C}_m, x, \tau_m) = \begin{cases} 1, & \text{if } |D^x - \tau_m| - \beta \cdot E_{\mathcal{C},x} < \delta_m \\ 0, & \text{otherwise} \end{cases}, \quad (7)$$

$$g_h(\mathcal{C}_h, x, \tau_h) = \begin{cases} 1, & \text{if } D^x > \tau_h \text{ and} \\ & D^x + \beta \cdot E_{\mathcal{C},x} > \delta_h, \\ 0, & \text{otherwise} \end{cases}, \quad (8)$$

where τ_e , τ_m , and τ_h are adaptive thresholds for each cache, and β is a coefficient controlling expressiveness. δ_e , δ_m , and δ_h represent the maximum difficulty in the easy cache, the maximum deviation from τ_m in the medium cache, and the minimum difficulty in the hard cache, respectively.

Specifically, a sample x with difficulty $D^x < \tau_e$ that exceeds the maximum difficulty of the easy cache δ_e would normally be excluded from $\mathcal{C}_e$. However, since diversity is a critical factor in cache construction, storing x can be more beneficial than discarding it if $E_{\mathcal{C},x}$ is sufficiently high. Therefore, we moderate the acceptance criteria by adjusting the comparison between δ_e and D^x by the magnitude of $\beta \cdot E_{\mathcal{C},x}$. Similarly, for the hard cache, samples satisfying $\tau_h < D^x < \delta_h$ with sufficient diversity replace existing

samples that have the minimum difficulty. The same principle applies to the medium cache.

Since evaluating all incoming data for caching introduces high computational overhead, we randomly sample a subset of $\mathcal{X}_t$ as candidates for caches updates. Due to the compact size of caches, this sampling ratio can be as low as 10% of the original dataset, significantly reducing the computational cost of cache maintenance. Our design, which integrates both the difficulty and expressiveness of incoming data enables the model to learn from more diverse samples even under the data-constrained conditions of cache-based training. Algorithm 1 presents the simplified procedure for sample selection in pseudocode.

3.2. Curricular Domain Incremental Training

3.2.1. Curriculum training

Building upon our difficulty-aware caches design for efficient domain-incremental learning, we adopt a curriculum training strategy [1] to enhance convergence and improve generalizability. Standard curriculum training involves a difficulty function to assess sample hardness and a pace function to determine the sampling strategy. We leverage our difficulty-aware caches to implement an easy-to-hard sampling strategy. Specifically, our pace function follows a cosine-based schedule that initially prioritizes the easy cache $\mathcal{C}_e$ and gradually increases the sampling probability from the hard cache $\mathcal{C}_h$ as training progresses. The sampling for the medium cache $\mathcal{C}_m$ peaks at the midpoint of the training process. To further stabilize learning, we incorporate cache-agnostic sampling phases at both the beginning and the end of training. In addition to the cache-based sampling schedule, we apply a progressive augmentation strategy where the intensity of data augmentation increases over time, further enhancing generalization by introducing more challenging examples in later stages.

3.2.2. Loss function

To facilitate the compact retention of prior knowledge and promote semantic diversity in the new domain, we adopt a contrastive learning objective alongside the conventional cross-entropy loss. This strategy applies a pull operation to samples in the easy cache and push operation to those in the hard cache. The medium cache is excluded from this contrastive objective, as its primary role is to bridge the transition between the easy and hard caches. The contrastive loss is formulated as:

$$\mathcal{L}_{cont} = \mathcal{L}_{pull}(Z_e) + \mathcal{L}_{push}(Z_h, m), \quad (9)$$

where Z_e and Z_h are sets of features extracted from the easy/hard caches and m denotes the margin.

The overall training objective is defined as:

$$\mathcal{L}_{all} = \mathcal{L}_{CE} + \gamma \cdot \mathcal{L}_{cont}, \quad (10)$$

Backbone	Train	Test			Avg.
		FF++	DF(FF)	DF-UN	
Xception	FF++	96.49	83.25	69.03	82.92
	DF(FF)	86.73	99.24	71.54	85.84
ConvNeXt-tiny	FF++	97.11	83.19	68.02	82.77
	DF(FF)	90.63	98.69	69.68	86.33
CLIP-ViT-B/16	FF++	95.86	80.51	73.53	83.30
	DF(FF)	78.86	98.95	79.43	85.75
SwinV2-T	FF++	97.64	81.74	72.06	83.81
	DF(FF)	89.72	99.39	73.65	87.59
Ours					
Xception	DA-cache	92.80	96.24	75.27	88.10
ConvNeXt		95.74	97.99	77.27	90.33
CLIP		91.93	97.50	77.49	88.97
SwinV2		94.38	98.76	76.54	89.89

Table 1. Image-level performance. FF++: FaceForensics++, DF(FF): DF40 known set, DF-UN: DF40 unknown set, DA-cache: ours. Shaded cells indicate in-domain results.

where $\mathcal{L}_{CE}$ is the cross-entropy loss and γ is a weighting coefficient for the contrastive loss.

By integrating these components, our framework improves performance while utilizing only a fraction of the data required for full-set training. Furthermore, our approach is well-suited for efficient domain-incremental training as it does not require domain labels for incoming data, which is a natural assumption for real-world scenarios.

4. Experiment

4.1. Setup

4.1.1. Dataset

Experiments were conducted using DF40 [54] dataset, supplemented by additional data generated via Veo 3 [6] and Kling 2.5 [22]. DF40 training set, derived from FaceForensics++ (FF++) [41] real images, was used for model training, while its corresponding test set was reserved for evaluation. The additional generative data was partitioned into an 80/20 train-test split.

DF40 is a comprehensive benchmark comprising over one million images across 40 manipulation methods. It incorporates recent generation techniques such as Stable Diffusion [40] and MidJourney [32]. Within the dataset, 31 manipulation methods are categorized as known-domain data for training, while the remaining methods are designated as unknown-domain for testing purpose only. Known-domain samples were synthesized from real images in either FaceForensics++ or Celeb-DF-v2. In contrast, the unknown set includes methods such as DeepFaceLab [36], HeyGen [14], MidJourney-6 [32], WhichIsReal [52], StarGAN [3], StarGANv2 [4], StyleCLIP [35], e4e [47], and CollabDiff [17]. Manipulation methods are further classified by type: Face-



Figure 3. Sample frames generated by the latest commercial video generation models. Same prompts were used for both videos.

Swapping (FS), Face-Reenactment (FR), Entire Face Synthesis (EFS), and Face Editing (FE). While the known-domain data consists of FS, FR, and EFS, the unknown-domain includes FE methods, which present larger domain gap relative to the training distribution. The supplementary dataset includes 100 videos from Veo 3 and 60 videos from Kling 2.5. We allocated 80 Veo and 48 Kling videos for the training set, with the remaining 20 and 12 videos utilized for testing. Figure 3 shows samples generated by these models.

4.1.2. Implement details

Each cache holds 1% of the original training set across all manipulation types (e.g., face swap, reenactment, full-face synthesis). Therefore, 3% of the total dataset is utilized for training. The difficulty balancing coefficient α is set to 0.5, with both entropy H and feature distance d are normalized to the range $[0, 1]$. The expressiveness control parameter β and the contrastive loss weight γ are both set to 0.1, with a margin of 0.5 applied to the contrastive loss. Samples are randomly selected during the cache construction phase. We employ a sampling ratio of 0.1, which significantly reduces computational overhead. Cache construction and curriculum training alternate every epoch. All models are trained for 10 epochs using the Adam optimizer with a batch size of 64 on a single NVIDIA A5000. The learning rate is set to 1×10^{-5} for CLIP-ViT/16 [38], while 5×10^{-5} is used for training Xception [5], ConvNeXt-Tiny [29], and SwinV2-T [28]. Unless specified, we report the image-level AUC as the primary evaluation metric. For other hyperparameter settings, refer to [54].

4.2. Results

4.2.1. Overall Comparison across Backbone Models

Table 1 presents a quantitative comparison of the proposed method across Xception, ConvNeXt-Tiny, CLIP-ViT-B/32, and SwinV2-T as a backbone network. In-domain results are highlighted in blue. Our method consistently improves performance across all architectures, achieving up to a 5.67

Backbone	Train	Test									Avg.
		DFLab	HeyGen	MJ6	WIR	StarG	StarG2	StCLIP	e4e	CDiff	
Xception	FF++	73.48	68.22	58.04	54.78	67.94	42.90	83.89	88.48	83.58	69.03
	DF-FF	86.67	71.07	40.68	79.64	60.21	64.15	72.52	85.90	83.02	71.54
	DA-cache	84.6	77.1	29.7	69.96	78.01	62.92	90.56	91.53	93.06	75.27
ConvNeXt-tiny	FF++	78.89	61.45	72.62	40.46	70.45	43.59	81.90	87.50	75.36	68.02
	DF-FF	88.47	73.11	36.70	71.20	70.15	64.20	67.62	80.62	75.04	69.68
	DA-cache	92.36	74.85	59.41	66.19	77.91	55.84	88.60	92.84	87.45	77.27
CLIP-ViT-B/16	FF++	71.84	75.26	82.36	50.31	72.73	44.04	93.87	88.50	82.86	73.53
	DF-FF	88.19	71.09	57.67	63.06	86.41	75.34	88.31	87.32	97.51	79.43
	DA-cache	87.74	75.68	72.80	55.09	86.66	61.48	82.91	91.84	83.25	77.49
SwinV2-T	FF++	82.97	80.47	79.06	51.18	76.06	50.42	78.37	75.74	74.29	72.06
	DF-FF	93.58	66.07	43.53	69.47	76.71	56.62	81.93	83.82	91.15	73.65
	DA-cache	92.51	78.65	46.36	62.35	78.48	60.41	94.42	94.61	81.07	76.54

Table 2. Image-level AUC on DF40 *unknown* set. Abbreviations in test set: DeepFaceLab (*DFLab*), MidJourney6 (*MJ6*), WhichisReal (*WIR*), StarGAN (*StarG*), StarGANv2 (*StarG2*), StyleCLIP (*StCLIP*), and CollabDiff (*CDiff*).

Method	Train	Test			
		FF++	FS	FR	EFS+VK
Domain incremental training setting					
Xception	FF++	96.49	84.83	82.63	77.07
	EFS+VK	63.87	65.80	72.64	99.82
ConvNeXt	FF++	97.11	85.75	82.68	79.49
	EFS+VK	62.45	66.96	72.54	99.81
CLIP	FF++	95.86	81.87	80.37	77.12
	EFS+VK	54.98	53.66	73.70	99.75
SwinV2	FF++	97.64	85.35	81.51	74.32
	EFS+VK	60.38	66.69	82.41	99.77
Ours					
Xception	DA-cache	91.18	83.31	81.20	95.35
ConvNeXt		93.35	86.66	84.87	98.43
CLIP		73.17	75.45	79.42	96.99
SwinV2		94.08	83.87	86.55	98.11

Table 3. Image-level performance including Veo 3 [6] and Kling 2.5 [22]. EFS+VK represents EFS group data combined with data generated through Veo 3 and Kling 2.5. Our results are trained with proposed cache-based framework given EFS+VK set, on top of FF++ pretrained weight. Shaded cells indicate in-domain results.

point gain across the pretraining, fine-tuning, and unseen domains, despite utilizing only 3% of the additional data. As shown in Table 1, naive training on the new domain leads to a significant performance decrease on the pretraining data. In contrast, our approach effectively mitigates catastrophic forgetting with minimal impact on in-domain performance while enhancing generalizability in most cases. Although a slight decrease in generalizability is observed for CLIP-ViT-B/16, the proposed method still prevents severe performance drop in the pretraining domain.

4.2.2. Generalizability on Unseen Domains

Table 2 provides a detailed performance on the DF40 unknown set, reporting AUC scores across various generative

Method	Train	Test			
		FF++	FS	FR	EFS+VK
Continual training setting					
DFIL	task 1	93.09	76.13	70.96	70.80
	task 2	82.91	83.37	79.52	73.27
	task 3	81.09	82.13	78.40	71.04
	task 4	78.75	82.31	78.50	72.20
CoReD	task 1	97.92	84.78	82.07	79.00
	task 2	92.44	89.09	78.28	73.31
	task 3	86.64	85.41	93.10	69.31
	task 4	90.79	81.48	82.59	89.29
Ours	task 1	95.94	83.42	77.81	75.70
	task 2	88.13	96.60	81.29	83.88
	task 3	79.07	92.99	98.93	84.51
	task 4	80.43	87.19	92.08	96.72

Table 4. Image-level performance including Veo 3 and Kling 2.5, evaluated in a continual training manner to show our framework can effectively learn from in-domain data while preserving previous information. Task 1 to task 4 corresponds to incremental learning using FF++, FS, FR, and EFS+VK, respectively.

methods. These results demonstrate that our curriculum-based domain-incremental training enhances generalization under data-limited settings and functions effectively without domain labels during training. This improvement is consistent across diverse backbone networks. Training on the full DF-FF dataset underperforms relative to our method, except for the CLIP model. Note that our approach improves generalizability performance by 3.96 while using only a fraction of the available data.

4.2.3. Incremental and Continual Learning

We further evaluate our framework against existing continual learning methods for deepfake detection, DFIL [34] and CoReD [19]. As shown in Table 4, our method outperforms in-domain performance across tasks 2, 3, and 4 by a substantial margin compared to the initial performance in task 1. Our framework demonstrates robust adaptability to new

Training components	Samples	AUC
Full set training	343K	87.59
Random set training	34.4K	89.78 $\pm$ 0.32
Cache training		89.18
+ expressiveness		89.76
+ curriculum training	19.7K	89.90
+ contrastive loss		90.12 $\pm$ 0.24

Table 5. Ablation study of the proposed framework using SwinV2-T. We compare full-set training, fixed-size random subset training, and our difficulty-aware cache training. For random subset training and our approach, we report the mean $\pm$ std over five runs.

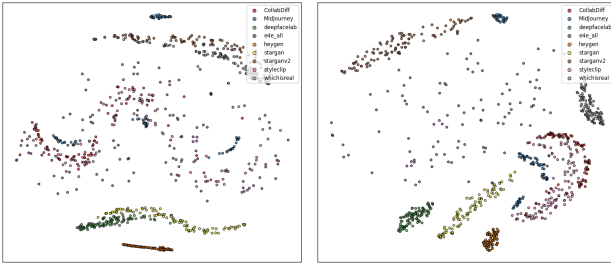


Figure 4. Visualization of t-SNE plot using features of model trained using DF-FF full set (left) and cache (right).

domains with large domain discrepancy, as shown in the final column. Regarding the preservation of prior knowledge, our approach shows an average performance degradation of 10.59 points across FF++, FS, and FR in-domain sets at task 4, while [19] shows 8.42. However, it is important to note that our method maintains a significantly higher in-domain performance; thus, the magnitude of the drop should be interpreted within the overall higher baseline.

4.2.4. Ablation study.

Table 5 presents an ablation study investigating the key components of our framework using SwinV2-T as the backbone. Full set training, where the model is trained on the entire DF-FF dataset, serves as the baseline. Random set training utilizes a fixed-size random subset of DF-FF to match the number of unique samples per epoch used in our method. In this analysis, expressiveness refers to the use of cosine similarity to ensure sample diversity during cache construction. Curriculum learning denotes the easy-to-hard sampling strategy applied to the caches, and contrastive loss indicates the inclusion of supervised contrastive losses applied to the easy and hard caches. While training with a random subset improves performance over full-set baseline, the cumulative number of unique samples encountered during the entire training process is $1.74\times$ larger than in our approach, as shown in the middle column. This underscores the efficiency of our framework, which achieves superior performance with a substantially smaller training data.

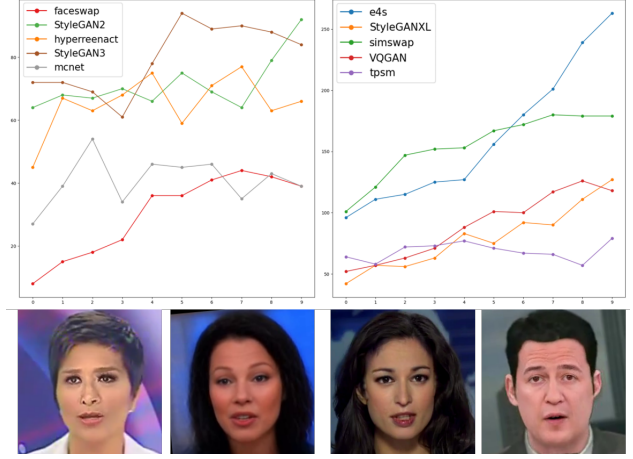


Figure 5. Number of images by forgery type in caches during training. Left: easy cache, Right: hard cache.

4.2.5. Analysis.

Figure 4 shows a t-SNE visualization of features from DF-UN images, extracted by models trained on the DF-FF full set and our proposed framework. The model trained on the full set fails to distinctively cluster images generated by different forgery methods. In contrast, our framework produces well-separated clusters for *DeepFaceLab*, *StarGAN*, *WhichisReal*, and others.

Figure 5 illustrates the evolution of sample distributions per forgery method within the caches across training epochs. We observe that the prevalence of specific forgery types within each cache increases over time, suggesting that the perceived difficulty of certain samples evolves as the training progresses. Our difficulty-aware caching and curriculum training effectively identify and leverage these evolving samples to preserve prior knowledge while continuously adapting to new domain challenges.

5. Conclusion

This paper presents a novel framework for domain-incremental continual learning in deepfake detection, centered on difficulty-aware caching and curriculum learning. By leveraging a compact subset of data, our design enables efficient adaptation to emerging domains while effectively mitigating the forgetting of prior knowledge. Compared to full-dataset training, the proposed method significantly reduces performance degradation and computational overhead. Furthermore, the integration of difficulty-aware caches with an easy-to-hard curriculum sampling strategy facilitates faster model convergence and enhances generalizability across unseen forgery methods. Experiments on recent deepfake benchmarks demonstrate the effectiveness of our framework, achieving a balance between adaptation and retention.

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